Q8. Autopsy case creation and disk-image analysis

Original question: Create a forensic case using Autopsy and analyze a disk image. Examine available files and metadata, identify and interpret at least two relevant digital artifacts.

Aim

Create a case, add the supplied image, inspect files and metadata, and identify at least two evidence-backed artifacts.

Requirements

Supplied image and all segments, known format/hash if available, writable case directory, adequate storage and Autopsy. No image was supplied here. References: [R5], [R6], [R7].

Procedure

Step 1 - Prepare input and select your Autopsy edition

```
mkdir -p "$HOME/EHF_Exam/Q8"
cd "$HOME/EHF_Exam/Q8"
IMAGE='<DISK_IMAGE_PATH>'
stat -c '%n : %s bytes' "$IMAGE"
sha256sum "$IMAGE" | tee image_before.sha256
```

Use a verified working copy. For segmented images, record/hash every segment and retain their names and order. Distinguish an E01 container hash from its raw acquisition digest.

Modern Autopsy 4.x has a desktop case/ingest wizard. Kali's `autopsy` package is the legacy browser-based 2.x edition. Use the matching workflow below. For modern Autopsy, use the lab's prepared desktop installation; `apt install autopsy` installs the legacy edition.

Step 2A - Modern Autopsy desktop workflow

1. Launch Autopsy. Choose New Case; enter EHF_Q8, a separate base directory and Single-user case when offered. Enter the supplied case number and your own examiner name; click Finish.
2. In Add Data Source, choose Disk Image or VM File. Browse to the supplied image/first supported segment. Set the evidence timezone according to the case facts; do not guess. Enter supplied hashes if the version offers this field.
3. Select relevant ingest modules: File Type Identification, Hash Lookup with a configured known-hash set, Keyword Search and relevant Recent Activity/Email modules where available. Record which were enabled; a hash-lookup module without a database cannot provide meaningful known-set matches.
4. Click Next/Finish and wait for ingest completion. Inspect task notifications for read errors or module failures; do not treat partial ingest as complete.
5. Expand Data Sources and the volume/file tree. Select a file; inspect the content viewer's File Metadata, Text and Hex views as applicable. Record its full evidence path, metadata identifier, size, allocation status and timestamps with timezone.
6. Expand Results/Data Artifacts (labels vary). Review a relevant parsed artifact such as a browser-history entry or email, if present. Inspect its source file and fields. Tag or bookmark the item and add a factual comment; do not infer behavior solely from its existence.

Step 2B - Legacy Kali browser workflow

1. Run `autopsy` in a terminal; follow the localhost URL printed by the installed version. If it explicitly requires elevated startup in your lab, follow that configured procedure. Keep the service local. Use the case evidence locker configured by the examiner.
2. Click New Case, enter case name, description and investigator, then New Host/Add Host. Record timezone and any clock adjustment provided in the scenario.
3. Click Add Image/Add Image File. Enter the absolute image path, select disk or partition as appropriate, and choose the available import/link mode according to the lab's preservation procedure. Review detected volumes and verify hashes through the supported integrity option or external SHA-256.
4. Open the volume and use File Analysis to browse directories. Select a file to inspect metadata and view text/hex. Use Keyword Search for a second relevant item if supported. Legacy releases do not provide the modern Recent Activity ingest tree.
5. Save notes/reports and screenshots from the available views. Label the edition in your practical record so GUI differences are explained.

Step 3 - Document two actual artifacts and verify

Choose two items actually present: a file plus a parsed browser record, two relevant files with metadata, or another supported combination. Do not count two views of one file as two independent artifacts.

```
sha256sum -c image_before.sha256
```

If files are exported through the GUI, save them outside evidence, hash each exported file and record its source path/identifier. Do not execute it.

Expected output - illustrative artifact worksheet

```
A1 | File artifact | path: [actual evidence path]
   | ID/size/times/status: [from file metadata]
A2 | Parsed record or second file | location: [actual source]
   | Relevant fields: [literal values observed]
```

Brackets are blanks to fill, not findings. Available artifacts depend on the image; verify case processing completed.

Interpretation, observation and result

Explain why A1 and A2 matter to the scenario and cite their source locations. A browser URL may indicate a stored history record, not that a particular person visited it; timestamps require timezone and context. Record "not available" for absent categories. The result is a created case and two supported artifacts if present; if fewer are present, state that limitation explicitly.

Common errors

- No Add Data Source wizard: check whether you are using legacy Autopsy; follow Step 2B.
- Unsupported/encrypted image: check format, segments and supported parsers; obtain the lab-approved decrypted copy/key procedure.
- Ingest incomplete: review module errors, storage and progress before reporting no artifacts.

Viva questions

- What is a case? A container for evidence references, analysis results and investigator notes.
- What is ingest? Automated processing that identifies files and extracts supported artifact types.
- Why preserve the image hash? To detect byte changes against the reference.
- Can metadata alone identify the actor? No; attribution needs corroborating evidence.